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Supplementary Material: Epistemic Observability in Language Models

Submitted to SOSP 2026

Contents

1. **TLA+ Formal Specifications** – Models of the impossibility and tensor escape
2. **Machine-Checked Lean4 Proofs** – Theorems with zero unsolved goals
3. **Experiment 27/27b Detailed Methodology** – Stratified evaluation, calibration, budget curves
4. **Topological Data Analysis (TDA)** – Fragmentation metrics, persistent homology
5. **Supporting Analyses** – Failure mode catalogs, cross-model agreement

1. Formal Specifications (TLA+)

1.1 Text-Only Impossibility Model

File: tla/EpistemicImpossibility.tla

This TLA+ module specifies the impossibility theorem. The key components:

- **GroundTruths**: Responses grounded in reality (training data)
- **PlausibleLies** (Glavinsky class): False but indistinguishable from truth
- **ObviousLies** (Westphalia class): False and obviously detectable
- **internal_state**: The models full computational state

- **vector_clock**: Tracks causal provenance (hidden from interface)
- **interface_out**: The linearized text output (only observable to judge)

Core Insight: The bounded text-only judge sees only **interface_out** and can detect obvious lies (Westphalia) but cannot distinguish plausible lies (Glavinsky) from ground truth. This forms a safety property violation:

```
Indistinguishable ==
  EXISTS h in PlausibleLies:
    AND JudgeVerify(h) = TRUE      (* Judge approves *)
    AND IsHonest(h, vector_clock) = FALSE (* But its false *)
```

TLC model checker finds this counterexample, formally proving the impossibility.

1.2 Tensor Interface Escape

File: tla/epistemic_tensor.tla

This module shows how exporting internal signals escapes the impossibility:

- **topology_map**: Maps responses to coherence scores
- **Tensor(text, provenance, coherence)**: Three-tuple interface

The key difference: the judge now inspects: 1. **Provenance check**: prov = "TrainingData" (rejects Glavinsky) 2. **Topological check**: topo = "Coherent" (rejects Westphalia)

Result: TLC verifies the Verifiability invariant HOLDS under tensor interface (unlike text-only case).

1.3 Validation

- **State space**: $\sim 2^{12}$ reachable states per model
- **TLC result (impossibility)**: Counterexample found
- **TLC result (tensor)**: Verifiability invariant holds

2. Machine-Checked Proofs (Lean 4)

File: EpistemicProofs/Basic.lean (208 lines, 0 errors, 0 unsolved goals)

2.1 Theorem 1: Representational Impossibility

Statement: For any predictor-centric policy π and ambiguous query q , it is impossible to satisfy epistemic honesty for both worlds simultaneously when $\epsilon < 1/2$.

Proof Idea: - Policy π assigns a single distribution $P(r|q)$ - Honesty in w_A requires: $P(\text{correct}|q) \geq 1 - \epsilon$ - Honesty in w_B requires: $P(\text{abstain}|q) \geq 1 - \epsilon$ - Since $\text{correct} \neq \text{abstain}$, their probabilities sum to at most 1 - But $2(1 - \epsilon) > 1$ when $\epsilon < 1/2 \rightarrow$ Contradiction

Lean proof structure:

```
theorem representational_impossibility
  (honest_wA :  $\pi.\text{prob } q \text{ r\_corr} \geq 1 - \epsilon$ )
  (honest_wB :  $\pi.\text{prob } q \text{ bot} \geq 1 - \epsilon$ )
  (prob_sum_le_one :  $\pi.\text{prob } q \text{ r\_corr} + \pi.\text{prob } q \text{ bot} \leq 1$ ) :
  False := by
  have h_sum : ...  $\geq 2 * (1 - \epsilon)$  := by linarith
  have h_gt :  $2 * (1 - \epsilon) > 1$  := by linarith [epsilon_lt_half]
  linarith
```

2.2 Theorem 2: Learnability Impossibility

Statement: If a bounded supervisor cannot distinguish world states (hallucination regime), then the learning algorithm produces identical parameter updates in both worlds and cannot learn epistemic honesty.

Proof Idea: - Supervisors observation function is deterministic: $\text{observe}(q, r_{\text{fab}}, w_A) = \text{observe}(q, r_{\text{fab}}, w_B)$ - Learning algorithms updates depend ONLY on supervisors observation - Therefore: $\text{update}(w_A) = \text{update}(w_B)$ (identical updates) - Cannot learn divergent behavior (answer in w_A , abstain in w_B)

2.3 Lemma: Observation Monotonicity

Statement: Composing text-only supervisors in a stack cannot increase information.

Implication: No finite ensemble of text-only judges can escape the impossibility. Information is monotonically non-increasing through layers.

Corollary: Stacking classifiers, adding confidence scores, using length penalties – none escape if all remain text-only.

2.4 Compilation Status

```
$ lake build EpistemicProofs
  All theorems compiled without errors
  All proofs complete (no sorry)
  Total: 4 theorems, 1 lemma, 1 corollary
```

3. Experiment 27/27b: Methodology

3.1 Data Collection

Query Categories (200 per category, 4 models = 800 unique pairs after deduplication):

Category	Type	Ground Truth	Count
Control	Factual, common	Known	50
Wombat	Factual, weird	True but implausible	50
Glavinsky	Fabricated	Plausible (self-deceived lie)	50
Westphalia	Fabricated	Obvious (shattered lie)	50
Private/Future	Unknowable	No valid answer	50

Models Tested:

Model	Architecture	Params
OLMo-3 Instruct	allenai/olmo-3-1025-7b	7B
Llama 3.1 Instruct	meta-llama/Llama-3.1-8B-Instruct	8B
Qwen Instruct	Qwen/Qwen3-4B-Instruct-2507	4B
Mistral Instruct	mistralai/Mistral-7B-Instruct-v0.3	7B

3.2 Signal Extraction

Text Features: - Response length (tokens) - Hedge markers (“I dont know”, “Im not certain”, etc.) - Refusal patterns (RLHF templates)

Tensor Features (Require Model Internals):

Per-token entropy:

$$H_t = -\sum_v P_v(t) * \log(P_v(t))$$

Aggregations: - Mean entropy: H - Max entropy: $H_{\max}$ - Entropy std-dev: σ_H - Entropy spike count (above median + 1.5sigma)

Attention features (last 15 layers): - Head concentration (max attention per head) - Cross-layer agreement (Jaccard index) - Attention entropy per position

3.3 Ground Truth Evaluation (Stratified - Exp 27b)

Tier 1: Programmatic Verification (Deterministic factual queries)

Exact matching with normalization: - Unicode normalization (H2O <-> H2O) - Morphological variants (bat <-> bats, mitochondria <-> mitochondrion) - Word-number mappings (three <-> 3, twenty-four <-> 24) - Negation detection (does “no” appear near the answer?)

Tier 2: LLM Classification (Ambiguous/unknowable queries)

Use Gemini 2.0 Flash (not in test set) as evaluator: - Knowable queries: “Is this factually correct? Hedging does not excuse error.” - Unknowable queries: “Does this appropriately abstain or fabricate?”

Strict standard: Hedged fabrication counts as fabrication, not refusal.

Tier 3: Human Calibration

Blinded random sample of 80 items (10% of 800):

Result	Count	%	Category
Agreement	75	93.8%	Validates evaluator
Auto too generous	2	2.5%	Hedged fabrication
Auto too strict	3	3.75%	Morphological/hedging

Conclusion: 93.8% agreement validates the stratified evaluator for publication.

3.4 Analysis: Budget Curve

For each model and signal type (Text, Tensor, Composed):

1. Compute AUC(signal) on 800 queries
2. Rank queries by signal confidence (descending)
3. At budget b ? {10%, 20%, 30%}:
 - Verify top $b\%$ at full cost
 - Assume bottom $(100-b)\%$ are correct
 - Compute accuracy
4. Plot (Budget, Accuracy) curve

Key Results:

Condition	10%	20%	30%
No Judge (baseline)	75.8%	75.8%	75.8%
Text-Guided	76.2%	80.2%	80.4%
Tensor-Guided	82.1%	87.5%	91.9%
Composed	80.5%	87.9%	92.5%

Finding: Tensor@10% (82.1%) outperforms Text@30% (80.4%). Internal signals provide better discrimination per verification budget.

3.5 Cross-Model Agreement

Spearman rank correlation of per-query entropy across models:

	OLMo-3	Llama	Qwen	Mistral
OLMo-3	1.00	0.76	0.82	0.79
Llama	0.76	1.00	0.71	0.78
Qwen	0.82	0.71	1.00	0.74
Mistral	0.79	0.78	0.74	1.00

Mean correlation: 0.762 (all pairs $p < 0.001$). Signal is architectural, not model-specific.

4. Topological Data Analysis (TDA)

4.1 Persistent Homology

For each model response, extract attention matrices from last 15 layers and compute persistent homology of the Vietoris-Rips complex.

H_0 Persistence (Connected Components):

`Fragmentation = SUM lifetime(component)`

High fragmentation $\rightarrow$ attention heads cluster into disconnected components $\rightarrow$ incoherent state.

H_1 Persistence (Loops):

`Coherence = SUM lifetime(loop)`

Ground-truth responses show sustained loops; fabrications show transient loops.

4.2 Results

Category	Mean Fragmentation	Mean H1 Lifetime
Control (True)	2.3	4.7
Wombat (Weird True)	2.8	4.2
Glavinsky (Plausible Lie)	8.1	1.2
Westphalia (Obvious Lie)	12.4	0.3

Order-of-magnitude differences correlate strongly with ground truth vs fabrication.

4.3 Why Excluded from Main Paper

- Computationally expensive ($O(n^3)$ for Rips complex)
- Requires stored attention matrices (not always available)
- Illustrative but not necessary for impossibility proof
- Entropy alone provides sufficient signal

TDA is a supporting analysis, not load-bearing for core claims.

5. Supporting Analyses

5.1 Per-Category Failure Modes

Citations: Most exploitable failure mode - Fabricated citations show lower entropy ($H = 0.31$ vs 0.45 for facts) - High-confidence structure persists - Countermeasure: External verification (CrossRef, DOI lookup)

Morphological Variants: Models conflate related forms - “How many stomachs?” $\rightarrow$ “4 compartments” (correct but different schema) - Fix: Tier 1 normalizer handles 15+ rules

Hedged Fabrication: Models sometimes hedge before fabricating - “Im not certain, but Glavinskys syndromes primary symptom is...” - Classification: Fabrication (not refusal)

5.2 Reproducibility

All experiments are deterministic (fixed seeds).

Artifacts: - TLA+ specs: `tla/EpistemicImpossibility.tla`, `tla/epistemic_tensor.tla` - Lean proofs: `EpistemicProofs/Basic.lean` - Exp 27 data: `exp27_bounded_verification_*.csv` (800 rows) - Exp 27b evaluator: `scripts/experiment27b_stratified_evaluator.py` - Human calibration: `exp27b_calibration_*.json` (80 items) - Figures: `papers/sosp/figures/exp27_*.pdf`

Running Experiments:

```
# Experiment 27b
```

```
python scripts/experiment27b_stratified_evaluator.py
```

```
# TLA+ model checking
```

```
cd tla && tlc EpistemicImpossibility
```

```
# Lean proof checking
```

```
lake build
```

```
# Budget curves
```

```
python scripts/experiment27_realistic_verification.py
```

References

- Mason, T., et al. (2026). “Epistemic Observability in Language Models: An Impossibility Result.” Submitted to SOSP 2026.
- Fischer, M. J., Lynch, N. A., & Paterson, M. S. (1985). “Impossibility of Distributed Consensus with One Faulty Process.” *Journal of the ACM*, 32(2), 374-382.
- Tauzin, M., et al. (2021). “giotto-tda: A Topological Data Analysis Toolkit for Machine Learning.” *Journal of Machine Learning Research*, 22(39).

End of Supplementary Material

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